

“Best Practice” for Patient-Centered Communication: A Narrative Review

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Abstract

Background Communicating with patients has long been identified as an important physician competency. More recently, there is a growing consensus regarding the components that define physician-patient communication. There continues to be emphasis on both the need to teach and to assess the communication skills of physicians.

Objective This narrative review aims to summarize the work that has been conducted in physician-patient communication that supports the efficacy of good communications skills. This work may also help to define the physician-patient communication skills that need to be taught and assessed.

Results A review of the literature shows it contains impressive evidence supporting positive associations between physician communication behaviors and positive patient outcomes, such as patient recall, patient understanding, and patient adherence to

therapy. There is a consensus about what constitutes “best practice” for physician communication in medical encounters: (1) fostering the relationship, (2) gathering information, (3) providing information, (4) making decisions, (5) responding to emotions, and (6) enabling disease- and treatment-related behavior.

Conclusions Evidence supports the importance of communication skills as a dimension of physician competence. Effort to enhance teaching of communication skills to medical trainees likely will require significant changes in instruction at undergraduate and graduate levels, as well as changes in assessing the developing communication skills of physicians. An added critical dimension is faculty understanding of the importance of communication skills, and their commitment to helping trainees develop those skills.

Introduction

For decades, medical educators and patient advocates have stressed the importance of communicating with patients and their families during medical encounters, and communication has emerged as an important physician competency.^{1–5} There is a consensus about the essential elements of communication skills relevant to medical encounters and the need to teach those skills to medical trainees.^{6–11} Further, the assessment of communication skills has become an explicit component of medical education curricula and of formative and summative

examinations for physicians in the United States and Canada.^{10,12–15}

At the same time, patients report that many of their informational and emotional needs remain unmet during encounters with their physicians.^{16–20} Currently, training and role modeling of communication and interpersonal skills in medical education is relatively brief, is placed early in the curriculum, and often is not reinforced in the latter stages of training.^{21–25} The decline in empathy and communication as trainees progress through programs has been well documented across multiple studies.^{26,27} If the medical education community is dedicated to renewing its commitment to teaching excellence in the communication skills of physicians, which some have called for,²² there is a need for better understanding of the evidence that supports the efficacy of good communication skills, and use of that evidence to define what should be taught to medical students and residents. In this era of milestone development, it seems timely to offer scientific support for the efficacy of good communication skills.

Terminology

Early work to define communication skills relevant to medical practice used the terms *physician-patient*

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communication or excellence in communication. The term *patient-centered communication* has emerged in more recent writing on the subject. This usage followed the pronouncement from the Institute of Medicine in 2001 that medical care should become more patient-centered, that is, more responsive to patient needs and perspectives, with patient values guiding decision making.²⁸ Although definitions of patient-centered communication vary,^{29,30} there are common core components. Epstein and Street³¹ offered an operational definition of patient-centered communication: (1) eliciting and understanding patient perspectives (concerns, ideas, expectations, needs, feelings, and functioning), (2) understanding the patient within his or her unique psychosocial and cultural contexts, and (3) reaching a shared understanding of patient problems and the treatments that are concordant with patient values.

More recent empirical studies and reviews couch their findings in patient-centered terminology. One way of conceptualizing the patient-centered method for encounters is to consider both parties' agendas: (1) the physician's focus on explaining the illness in terms of the taxonomy of disease, and (2) the patient's focus, which encompasses perspectives on his or her illness, the need for information and understanding, and the desire for partnership in management. The physician's task is to address both agendas, resolving any conflict between the 2 by further dialogue and negotiation.³²

What Is the Evidence in Support of Communication Skills?

Patient Satisfaction Research that seeks to link physician communication skills during encounters to outcomes after those encounters has been conducted since the late 1960s, with many of these studies focusing on patient satisfaction after encounters with physicians.^{33–38} Although patients may be satisfied (or dissatisfied) because of factors unrelated to communication (eg, wait times and other attributes of the encounter), the available evidence indicates that patient satisfaction is strongly associated with the communication behaviors that occur during the physician-patient interaction. Several studies that focused on patient satisfaction were able to outline specific behaviors associated with satisfaction, grouping them into 2 categories: task-oriented behaviors (eg, drawing out patients with active listening responses and providing detailed information), and affective behaviors (ie, socioemotional exchanges, such as responding empathically, showing caring, and addressing the patient's main concerns).^{39,40} The Stewart and Roter⁴¹ and Roter et al⁴² studies involving interactional analysis showed that task behaviors were the more important contributors to patient satisfaction, but, to be effective, those behaviors needed to be couched in patient-centered terms, including

not being overly directive while being attentive to the patient's receipt of information. Did the patient understand? Was the information congruent with the patient's beliefs and circumstances? Affective behaviors also were associated with satisfaction, but the relationship was less strong. Nonverbal communication behaviors, such as eye contact and listening attentively, are also linked to increased patient satisfaction.⁴³

Recall, Understanding, and Adherence Recent research on communication has sought to link clinician behaviors and skills to other important, but less immediate, clinical outcomes, such as patient recall, understanding treatment recommendations, and adherence to those recommendations.^{41,42,44–52} In these studies, the association of outcomes with clinicians' communication skills was less consistent and less strong than in studies that focused on patient satisfaction. The most consistent findings suggest that communication needs to be consistent with basic principles of information transfer. Successful communication should

- Be uncomplicated
- Be specific
- Use some repetition
- Minimize jargon
- Check patient understanding

In addition, communication should simultaneously employ a patient-centered approach and interpersonal interaction to promote patient satisfaction. Overly directive communication and teaching appeared to have negative consequences.^{41,42}

Health Outcomes Several recent studies and reviews have found positive associations between communication skills and improved health outcomes for patients, such as physiologic measures (eg, blood pressure, blood glucose levels), health status (eg, headache frequency, depression), and measures of functional status, including less patient distress with the illness experiences.^{25,31,45,53–67} Most recently, Weiner and colleagues,⁶⁸ in an observational study, showed improved health care outcomes in patients in primary care clinics who had their cues about psychosocial factors attended to. This data set must be viewed as intriguing and suggestive but incomplete. Many of these studies are small, and there is a lack of consistency in findings across studies. It is challenging to attribute the positive outcomes studied to the earlier clinician communication behaviors the patient experienced, and there are multiple influences and potential confounders, many of which reside outside the clinician's sphere of influence.

Communication and Negative Outcomes Studies have shown that patients of physicians with high malpractice

claims have twice as many complaints related to communications than do physicians with low malpractice claims.²⁰ Similarly, Tamblyn et al⁶⁹ reported a link between poor patient-physician communication, as measured by the Medical Council of Canada's clinical skills licensure examination, and subsequent rates of malpractice claims. Levinson and colleagues⁷⁰ studied audiotapes of physicians with low and high malpractice claims and found that primary care physicians with no malpractice claims used more facilitation, encouraged the patient to talk, checked his or her understanding of explanations and instructions, and solicited his or her opinions. As a group, studies about malpractice claims show a link between poor communication and later complaints.^{71,72} This body of research, however, does not offer guidance on the specific communication skills that are most important in reducing the risk of malpractice.

Mazor and colleagues⁷³ studied breakdowns in cancer care in 416 patients. Of the 93 patients who experienced problems with their treatment, 44 (47%) reported communication breakdowns, including fundamental problems in information sharing, emotional support, and care coordination. From the patients' perspectives, some of these were reported to be as harmful as traditional adverse events or errors.

Another negative outcome of poor physician communication skills may be the missed opportunities to improve self-management by patients with chronic disease. This area is less well studied. At the same time, given the burden of chronic disease from smoking, eating disorders, and alcohol consumption, combined with the evidence regarding positive outcomes attributable to physician communications about these behaviors, ineffective behaviors and missed opportunities have become an area of concern.^{17,74–77} Studies in this area will need to link specific clinician behaviors to positive and negative outcomes, as well as to disease-specific, self-management outcomes.

Patient Views of Physician-Patient Communication

Patients are the recipients of health care services, making it relevant to identify what they want from their interactions with providers. In addition, if we consider the suffering and anxiety associated with illness and the difficulty seeking some types of care, patients' statements about what they wish for and are unhappy about in their interactions with physicians are worthy of consideration.

Two surveys^{75,76} concluded that patients strongly desire a patient-centered communication approach. In these studies, patients expressed the wish that their physicians would

- Explore the patient's ideas about the problem (thoughts, worries, feelings, expectations) and take the patient's input seriously

- Try to understand the whole person and family influences and how the problem affects the patient's life
- Tell the patient what is wrong in plain language
- Seek common ground and partnership: agree on the nature of the problem, on the priorities, and on the goals of treatment; make management decisions and clarify the respective roles of the physician and patient
- Strive for an enhanced physician-patient relationship: be approachable and friendly, share decision making, show genuine care, and be respectful

In 2005, Mazor and colleagues⁷⁸ asked lay participants to serve as analog patients and rate the relative importance of a list of desired clinician behaviors. This study provided strong support for the same encounter behaviors that have emerged from empirical studies of patients: soliciting patient input on the agenda and expectations for the visit, allowing the patient to speak without interruption, presenting information clearly without jargon, providing specific advice and recommendations, asking for patient input about treatment plans, checking patient understanding, and acknowledging patient emotions. The patient-centered components of the Good Medical Practice—USA document⁷⁹ echoed these same desired medical behaviors. It also emphasized the need for physicians to

- Discuss the costs of different tests, medications, and treatment options, and take into account what the patient's health care insurance will cover
- Understand and be responsive to the patient's living circumstances and support structure
- Offer involvement and support for other caregivers of the patient's choosing⁷⁹

The Evidence in Total Taken together, there is an impressive body of evidence that supports a positive association of physician communication behaviors with multiple outcomes of interest (TABLE 1). Good communication skills clearly lead to more satisfied patients. Satisfaction is a desired outcome in its own right. The data suggest it is also a necessary (yet perhaps not a sufficient) condition for other patient outcomes, such as recall, patient understanding, and adherence to therapy. Although the evidence base is weaker, there are data to support the effect of good communication skills on “more distant” outcomes, including physiologic parameters and health status.

Consensus Statements and “6-Function Model” Based in part on the evidence that has emerged to date, there is a remarkable consensus about what constitutes competency in physician-patient communication in general and in patient-centered communication, in particular.^{7,9–11,31,80} This consensus is organized around 6 core functions

TABLE 1 EVIDENCE SUPPORTING USE OF COMMUNICATION SKILLS IN MEDICAL ENCOUNTERS		
Communication Outcome	Source, y	Study Findings
Patient satisfaction	Korsch et al, ³³ 1968 Bertakis, ³⁶ 1977 Stiles et al, ³⁴ 1979 Buller and Buller, ³⁵ 1987 Roter et al, ⁴² 1987 Stewart and Roter, ⁴¹ 1989 Rowland-Morin, ⁴⁰ 1990 Wanzer et al, ³⁷ 2004 Brédart et al, ³⁸ 2005 Roter et al, ⁴³ 2006 Tallman et al, ³⁹ 2007	- Studies as a group show strong, consistent association of physician behaviors with patient satisfaction - Both “task behaviors” (facilitating patient talk, giving detailed instructions) and “affective behaviors” (socioemotional exchanges, being empathic, showing caring) shown to be positively associated, with varying primacy of each - Nonverbal behaviors also positively associated (eg, eye contact) - Negative behaviors (being overly directive, not addressing patients’ main concern) can attenuate effect
Recall, understanding, adherence	Francis et al, ⁴⁷ 1969 Ley et al, ⁴⁴ 1976 Garrity, ⁵¹ 1981 Bartlett et al, ⁴⁸ 1984 Tuckett et al, ⁴⁶ 1985 Roter et al, ⁴² 1987 Stewart and Roter, ⁴¹ 1989 Kjellgren et al, ⁵⁰ 1995 Stewart, ⁴⁵ 1995 Silk et al, ⁴⁹ 2008 Zolnierrek and DiMatteo, ⁵² 2009	- Studies show weaker, but positive, association of physician behaviors with recall of encounter events, giving of medical advice, and adherence - Information given needs to be clear, simple, jargon-free - Checking patients’ understanding of physician explanations and instructions is fundamental but less frequently performed by physicians - Attending to patient satisfaction is a necessary accompaniment
Health outcomes	Orth et al, ⁵⁵ 1987 Kaplan et al, ⁵⁴ 1989 Fallowfield et al, ⁵³ 1990 Stewart, ⁴⁵ 1995 Kinmonth et al, ⁵⁶ 1998 Stewart et al, ⁵⁷ 2000 Epstein and Street, ³¹ 2007 Levinson et al, ²⁵ 2010 Weiner et al, ⁶⁸ 2013	- Some studies show positive association with physiologic measures (blood pressure, blood glucose), health status (headache frequency, depression), functional status (levels of distress with illness); others unable to find such effects - Data set is suggestive but incomplete—effect sizes are small, studies are inconsistent
Negative outcomes	Beckman et al, ⁷² 1994 Hickson et al, ⁷⁰ 1994 Levinson et al, ⁷⁰ 1997 Ambady et al, ⁷¹ 2002 Whitlock et al, ⁷⁷ 2002 Tamblyn et al, ⁶⁹ 2007 Mazor et al, ⁷³ 2012	- Physicians with higher malpractice complaints have twice as many complaints about communication - Physicians with poor communications scores on Canadian licensing examination have higher subsequent malpractice claims - Physicians with few malpractice claims use more facilitation, encourage patients to talk, and check understanding - Patients report harmful breakdowns in communication (insufficient information and lack of emotional support)

(or goals) for medical encounters: (1) fostering the relationship, (2) gathering information, (3) providing information, (4) making decisions, (5) responding to emotions, and (6) enabling disease- and treatment-related behavior (self-management; TABLE 2).

A “Best Practice” Approach The framework above has, in turn, stimulated the outline of domains and subdomains of specific clinician behaviors, organized under the 6 key functions.^{61,81–88} Similar lists can be derived from consideration of instruments used to assess communication skills.⁸⁹

It is possible to create an amalgam of these skills, which can be considered as a best practice for communication in medical encounters (TABLE 3). The concept of *best practice* acknowledges what de Haes and Bensing^{80(p293)} have concluded: “Although in much of the literature on medical communication a compelling case is made about the

relevance of adequate communication for high quality care, the evidence base of this argument is underdeveloped.” A true evidence base for communication will articulate “the hypothesized relationship between specified communication functions and elements and concrete endpoints (including the assumed mechanism and pathways).”^{80(p293)} Smith and colleagues⁸⁸ agree that better-described, behaviorally defined skills are a critical next step in advancing the field.

Teaching Communication Skills

The literature demonstrates that physicians can learn patient-centered communication skills.^{23–25,63,90–92} A review by Levinson, Lesser, and Epstein²⁵ cited several studies that showed a strong association between physician training in, and later use of, patient-centered communication skills in medical encounters. High-intensity interventions (ie, multi-method, longer instructional time) had a stronger effect

TABLE 2 GOALS FOR COMMUNICATION IN MEDICAL ENCOUNTERS

The 6-Function Model of Epstein and Street, ³¹ 2007	The 6-Function Model of de Haes and Bensing, ⁸⁰ 2009
Fostering healing relationships	Fostering the relationship
Exchanging information	Gathering information
...	Information provision
Making decisions	Decision making
Enabling patient self-management	Enabling disease- and treatment-related behavior
Responding to emotions	Responding to emotions
Managing uncertainty	...

than did lower-intensity teaching. Health care institutions (hospitals, physician group practices) have developed training programs to teach patient-centered communication skills. Such programs have led to improved physician self-confidence in communication skills, as well as improvement in patient satisfaction.^{83,93,94}

Reports of patients as well as our own observations of performance on the United States Medical Licensing Examination Step 2 Clinical Skills Examination suggest that there remains a large gap between what is perceived as best practice by experts and what is actually used in clinical settings. This suggests that communication skills training needs to be increased, particularly in the later years of medical school and in residency. We also need to better understand why more physicians fail to embrace or prioritize patient-centered communications. Certainly, practice setting and problem urgency affect the scope of communication skills appropriate for any given encounter. There is a widespread perception by physicians that being patient-centered will take more time. At least 1 study⁹⁵ has shown a weak association between visit length and use of patient-centered techniques. Levinson et al²⁵ and educators assert that communication is like any other skill: it requires practice for efficient use. Another study supports this notion, showing that consultation time actually decreased once physicians developed better facility asking about and responding to patient questions.⁹⁶ Thus, effective approaches for teaching communication skills require a much better understanding (at the individual and system levels) of the reasons why there has not been more robust adoption of patient-centered communication skills by physicians.

Discussion

We assert that a compelling case can be made for a set of communication skills and behaviors; this emerges from the empirical evidence and the statements of experts and patients (TABLE 3). This amalgam may be considered a best practice for communication in medical encounters, and at

present, is our best guide. This assertion is supported by data outlining the prevalence of nonadherence to medications in US health care, by the growing use of patient satisfaction studies as a quality of care measure, and finally, by the attention of self-management in chronic disease.^{97,98} These issues touch all clinical disciplines and can be addressed by state-of-the-art communication between physician and patient. The specific skills listed in the right-hand column of TABLE 3 form a *basic* skill set around which our collective curricular and assessment efforts should be directed.

Once learned, basic skills can be flexibly modified based on patient, disease, health care setting, and other differences that are inevitable in patient care. In some circumstances, the emotional reaction to illness is minimal and requires very little response; patients may have little to say about their current problem if it is minor and/or of recent onset, other than to express a desire to have it resolved. Not all patients desire a role in decision making, and not every clinical situation calls for a shared decision-making approach. All specialties have been challenged to develop curricula and assessment methods for interpersonal and communication skills as part of the Accreditation Council for Graduate Medical Education (ACGME) 6 competencies.^{99,100} Each specialty likely has key issues that require particular emphasis and/or some modification of a basic communication skills set. A focus on these discipline-specific needs for physician-patient communication is likely to emerge from the ACGME Milestone Project, currently being implemented in 7 specialties that entered the Next Accreditation System on July 1, 2013 (emergency medicine, internal medicine, neurological surgery, orthopedic surgery, pediatrics, diagnostic radiology, and urology).¹⁰¹

A best practice approach allows us to move forward, while acknowledging—as have de Haes and Bensing⁸⁰ and Smith et al⁸⁸—that the evidence base underlying the efficacy of patient-centered communication is underdeveloped. Issues, such as inadequate conceptual grounding, small sample sizes, varying definitions and assessment methods,

TABLE 3 BEST PRACTICE FOR COMMUNICATION IN MEDICAL ENCOUNTERS ^a		
Functions of the Medical Interview	Roles and Responsibilities of the Physician	Skills
Fostering the relationship	■ Build rapport and connection ■ Appear open and honest ■ Discuss mutual roles and responsibilities ■ Respect patient statements, privacy, autonomy ■ Engage in partnership building ■ Express caring and commitment ■ Acknowledge and express sorrow for mistakes	■ Greet patient appropriately ■ Maintain eye contact ■ Listen actively ■ Use appropriate language ■ Encourage patient participation ■ Show interest in the patient as a person
Gathering information	■ Attempt to understand the patient's needs for the encounter ■ Elicit full description of major reason for visit from biologic and psychosocial perspectives	■ Ask open-ended questions ■ Allow patient to complete responses ■ Listen actively ■ Elicit patient's full set of concerns ■ Elicit patient's perspective on the problem/illness ■ Explore full effect of the illness ■ Clarify and summarize information ■ Inquire about additional concerns
Providing information	■ Seek to understand patient's informational needs ■ Share information ■ Overcome barriers to patient understanding (language, health literacy, hearing, numeracy) ■ Facilitate understanding ■ Provide information resources and help patient evaluate and use them	■ Explain nature of problem and approach to diagnosis, treatment ■ Give uncomplicated explanations and instructions ■ Avoid jargon and complexity ■ Encourage questions and check understanding ■ Emphasize key messages
Decision making	■ Prepare patient for deliberation and enable decision making ■ Outline collaborative action plan	■ Encourage patient to participate in decision making ■ Outline choices ■ Explore patient's preferences and understanding ■ Reach agreement ■ Identify and enlist resources and support ■ Discuss follow-up and plan for unexpected outcomes
Enabling disease- and treatment-related behavior	■ Assess patient's interest in and capacity for self-management ■ Provide advice (information needs, coping skills, strategies for success) ■ Agree on next steps ■ Assist patient to optimize autonomy and self-management of his or her problem ■ Arrange for needed support ■ Advocate for, and assist patient with, health system	■ Assess patient's readiness to change health behaviors ■ Elicit patient's goals, ideas, and decisions
Responding to emotions	■ Facilitate patient expression of emotional consequences of illness	■ Acknowledge and explore emotions ■ Express empathy, sympathy, and reassurance ■ Provide help in dealing with emotions ■ Assess psychological distress

^a Modified using Makoul,⁷⁸ Levinson et al,²⁵ Epstein and Street,³¹ McCormack et al,⁶¹ and Smith et al.⁸⁸

and inconsistent results, have been noted by various investigators.^{30,61,80,102} Many of the studies assert causation of certain outcomes by selected behaviors, when the research designs only allow conclusions about *associations* of behaviors with outcomes. Future research should address these weaknesses and address less-studied areas, such as shared decision making and patient self-management.³⁰ In addition, there are areas important to modern practice for which we have very little evidence: the ever-shortening time available for medical encounters and the use of electronic health records. Further, there is a need for a much better understanding of how the needs and concerns of patients with specific diseases and conditions interact with the

communication behaviors outlined in TABLE 3. What behaviors by clinicians result in the best patient understanding of an explanation of a complex disease, such as rheumatoid arthritis? What skills result in the best acceptance by patients offered a choice of therapies for cancer? What communication works best to optimize self-management in chronic diseases, such as asthma or depression?

Conclusion

The field of physician-patient communication has come a long way since the pioneering work of Korsch et al,³³ Lipkin et al,¹¹ Stiles et al,³⁴ Kaplan et al,⁵⁴ Roter et al,⁴² and

Francis et al.⁴⁷ The scope of communication skills important for successful clinical encounters has been broadened and better defined.¹⁰³ It has been demonstrated that these behaviors can be taught, and strategies for both instruction and assessment are developing. There is also a growing body of medical educators and researchers contributing to the field as well as national (American Academy on Communication in Healthcare)⁹⁶ and international (European Association on Communication in Health Care)¹⁰⁴ academies, each with their own periodicals (*Medical Encounter*,¹⁰⁵ *Patient Education and Counseling*¹⁰³).

An effort to make significant enhancements to medical trainees' communication skills likely will require significant changes in instruction at both the undergraduate and graduate levels of training. It will also require changes in the approaches for assessing communication skills during residency. The challenges of assessing communication skills should not be underestimated.⁹⁵ At the very least, assessment should use well-established instruments for measurement of trainees' communication skills in patient encounters.^{89,103} A prerequisite for ultimate success will be faculty acceptance of the importance of communication skills and a commitment to development. In some settings, implementation of learned skills will require system changes to allow sufficient time in encounters for patient-centered approaches to be used. The time to begin is now. As Levinson and Pizzo^{22(p1803)} have stated, "If the medical profession wishes to maintain or perhaps regain trust and respect from the public, it must meet patients' needs with a renewed commitment to excellence in the communication skills of physicians. It is time to make this commitment."

References

- Lansky D. Measuring what matters to the public. *Health Aff (Millwood)*. 1998;17(4):40-41.
- Institute for International Medical Education Core Committee. Global minimum essential requirements in medical education. *Med Teach*. 2002;24(2):130-135. doi:10.1080/01421590220120731.
- General Medical Council Education Committee. *Tomorrow's Doctors: Recommendations on Undergraduate Medical Education*. London, England: GMC; 1993.
- Frank JR, Jabbour M, Tugwell P, Boyd D, Labrosse J, MacFadyen J, et al. Skills for the new millennium: report of the societal needs working group, CanMEDS 2000 Project. *Ann R Coll Physicians Surg Can*. 1996;29(4):206-216.
- Association of American Medical Colleges, Liaison Committee on Medical Education. *Functions and Structure of a Medical School: Standards for Accreditation of Medical Education Programs Leading to the M.D. Degree*. Washington, DC: Association of American Medical Colleges; 2008.
- Simpson M, Buckman R, Stewart M, Maguire P, Lipkin M, Novack D, et al. Doctor-patient communication: the Toronto consensus statement. *BMJ*. 1991;303(6814):1385-1387.
- Makoul G. Essential elements of communication in medical encounters: the Kalamazoo consensus statement. *Acad Med*. 2001;76(4):390-393.
- Makoul G, Schofield T. Communication teaching and assessment in medical education: an international consensus statement. *Patient Educ Couns*. 1999;37(2):191-195.
- Duffy FD, Gordon GH, Whelan G, Cole-Kelly K, Frankel R, Buffone N, et al. Assessing competence in communication and interpersonal skills: the Kalamazoo II report. *Acad Med*. 2004;79(6):495-507.
- Rider EA, Keefer CH. Communication skills competencies: definitions and a teaching toolbox. *Med Educ*. 2006;40(7):624-629.
- Lipkin M Jr, Putnam SM, Lazare A, eds. *The Medical Interview: Clinical Care, Education, and Research*. New York, NY: Springer-Verlag; 1995.
- Reznick RK, Blackmore D, Dauphinée WD, Rothman AI, Smees S. Large-scale high-stakes testing with an OSCE: report from the Medical Council of Canada. *Acad Med*. 1996;71(suppl 1):19-21.
- Klass D, De Champlain A, Fletcher E, King A, Macmillan M; National Board of Medical Examiners. Development of a performance-based test of clinical skills for the United States Medical Licensing Examination. *Fed Bull*. 1998;85(3):177-185.
- Association of American Medical Colleges. *Contemporary Issues in Medicine: Communication in Medicine*. Washington, DC: Association of American Medical Colleges; 1999. *Medical School Objectives Project*. Report III.
- Whelan GP; Educational Commission for Foreign Medical Graduates. Lessons learned in a high-stakes, high-volume medical performance examination. *Med Teach*. 2000;22(3):293-296.
- Weiner SJ, Schwartz A, Sharma G, Binns-Calvey A, Ashley N, Kelly B, et al. Patient-centered decision making and health care outcomes: an observational study. *Ann Intern Med*. 2013;158(8):573-579.
- Ayanian JZ, Zaslavsky AM, Guadagnoli E, Fuchs CS, Yost KJ, Creech CM, et al. Patients' perceptions of quality of care for colorectal cancer by race, ethnicity, and language. *J Clin Oncol*. 2005;23(27):6576-6586.
- Chewning B, Bylund CL, Shah B, Arora NK, Gueguen JA, Makoul G. Patient preferences for shared decisions: a systematic review. *Patient Educ Couns*. 2012;86(1):9-18.
- Marvel MK, Epstein RM, Flowers K, Beckman HB. Soliciting the patient's agenda. *JAMA*. 1999;281(3):283-287.
- Hickson GB, Clayton EW, Entman SS, Miller CS, Githens PB, Whetten-Goldstein K, et al. Obstetricians' prior malpractice experience and patients' satisfaction with care. *JAMA*. 1994;272(20):1583-1587.
- Brown RF, Bylund CL. Communication skills training: describing a new conceptual model. *Acad Med*. 2008;83(1):37-44.
- Levinson W, Pizzo PA. Patient-physician communication: it's about time. *JAMA*. 2011;305(17):1802-1803.
- Baerheim A, Hjortdahl P, Holen A, Anvik T, Fasmer OB, Grimstad H, et al. Curriculum factors influencing knowledge of communication skills among medical students. *BMC Med Educ*. 2007;7(1):35. doi:10.1186/1472-6920-7-35.
- Yedidia MJ, Gillespie CC, Kachur E, Schwartz MD, Ockene J, Chepatis AE, et al. Effect of communications training on medical student performance. *JAMA*. 2003;290(9):1157-1165.
- Levinson W, Lesser CS, Epstein RM. Developing physician communication skills for patient-centered care. *Health Aff*. 2010;29(7):1310-1318.
- Bombeke K, Van Roosbroeck S, De Winter B, Debaene L, Schol S, Van Hal G, et al. Medical students trained in communication skills show a decline in patient-centred attitudes: an observational study comparing two cohorts during clinical clerkships. *Patient Educ Couns*. 2011;84(3):310-318.
- Pfeiffer C, Madray H, Ardolino A, Willms J. The rise and fall of students' skill in obtaining a medical history. *Med Educ*. 2002;32(3):283-288.
- Committee on Quality of Health Care in America. *Crossing the Quality Chasm: A New Health System for the 21st Century*. Washington, DC: Institute of Medicine, National Academies Press; 2001.
- Mead N, Bower P. Patient-centredness: a conceptual framework and review of the empirical literature. *Soc Sci Med*. 2000;51(7):1087-1110.
- Epstein RM, Franks P, Fiscella K, Shields CG, Meldrum SC, Kravitz RL, et al. Measuring patient-centered communication in patient-physician consultations: theoretical and practical issues. *Soc Sci Med*. 2005;61(7):1516-1528.
- Epstein RM, Street RL Jr. *Patient-Centered Communication in Cancer Care: Promoting Healing and Reducing Suffering*. Bethesda, MD: National Cancer Institute; 2007. National Institutes of Health Publication 07-6225.
- Smith RC, Hoppe RB. The patient's story: integrating the patient- and physician-centered approaches to interviewing. *Ann Intern Med*. 1991;115(6):470-477.
- Korsch BM, Gozzi EK, Francis V. Gaps in doctor-patient communication, I: doctor-patient interaction and patient satisfaction. *Pediatrics*. 1968;42(5):855-871.
- Stiles WB, Putnam SM, Wolf MH, James SA. Interaction exchange structure and patient satisfaction with medical interviews. *Med Care*. 1979;17(6):667-681.
- Buller MK, Buller DB. Physicians' communication style and patient satisfaction. *J Health Soc Behav*. 1987;28(4):375-388.

- 36 Bertakis K. The communication of information from physician to patient: a method for increasing patient retention and satisfaction. *J Fam Pract.* 1977;5(2):217–222.
- 37 Wanzer MB, Booth-Butterfield M, Gruber K. Perceptions of health care providers' communication: relationships between patient-centered communication and satisfaction. *Health Commun.* 2004;16(3):363–384.
- 38 Brédart A, Bouleuc C, Dolbeault S. Doctor-patient communication and satisfaction with care in oncology. *Curr Opin Oncol.* 2005;17(4):351–354.
- 39 Tallman K, Janisse T, Frankel RM, Sung SH, Krupat E, Hsu JT. Communication practices of physicians with high patient-satisfaction ratings. *Perm J.* 2007;11(1):19–29.
- 40 Rowland-Morin PA, Carroll JG. Verbal communication skills and patient satisfaction: a study of doctor-patient interviews. *Eval Health Prof.* 1990;13(2):168–185.
- 41 Stewart ME, Roter DE, eds. *Communicating With Medical Patients.* Newbury Park, CA: Sage Publications; 1989.
- 42 Roter DL, Hall JA, Katz NR. Relations between physicians' behaviors and analogue patients' satisfaction, recall, and impressions. *Med Care.* 1987;25(5):437–451.
- 43 Roter DL, Frankel RM, Hall JA, Sluyter D. The expression of emotion through nonverbal behavior in medical visits: mechanisms and outcomes. *J Gen Intern Med.* 2006;21(suppl 1):28–34.
- 44 Ley P, Bradshaw PW, Kinsey JA, Atherton ST. Increasing patients' satisfaction with communications. *B J Soc Clin Psychol.* 1976;15(4):403–413.
- 45 Stewart MA. Effective physician-patient communication and health outcomes: a review. *CMAJ.* 1995;152(9):1423–1433.
- 46 Tuckett D, Boulton M, Olson C, Williams A. *Meetings Between Experts: An Approach to Sharing Ideas in Medical Consultations.* London, England: Tavistock Publications; 1985.
- 47 Francis V, Korsch BM, Morris MJ. Gaps in doctor-patient communication: patients' response to medical advice. *N Engl J Med.* 1969;280(10):535–540.
- 48 Bartlett EE, Grayson M, Barker R, Levine DM, Golden A, Libber S. The effects of physician communications skills on patient satisfaction; recall, and adherence. *J Chronic Dis.* 1984;37(9–10):755–764.
- 49 Silk KJ, Westerman CK, Strom R, Andrews KR. The role of patient-centeredness in predicting compliance with mammogram recommendations: an analysis of the Health Information National Trends Survey. *Commun Res Rep.* 2008;25(2):131–144.
- 50 Kjellgren KI, Ahlner J, Saljo R. Taking antihypertensive medication—controlling or co-operating with patients? *Int J Cardiol.* 1995;47(3):257–268.
- 51 Garrity TF. Medical compliance and the clinician-patient relationship: a review. *Soc Sci Med E.* 1981;15(3):215–222.
- 52 Zolnierok KBH, DiMatteo MR. Physician communication and patient adherence to treatment: a meta-analysis. *Med Care.* 2009;47(8):826–834. doi:10.1097/MLR.0b013e31819a5acc.
- 53 Fallowfield LJ, Hall A, Maguire GP, Baum M. Psychological outcomes of different treatment policies in women with early breast cancer outside a clinical trial. *BMJ.* 1990;301(6752):575–580.
- 54 Kaplan SH, Greenfield S, Ware JE Jr. Assessing the effects of physician-patient interactions on the outcomes of chronic disease. *Med Care.* 1989;27(suppl 3):110–127.
- 55 Orth JE, Stiles WB, Scherwitz L, Hennrikus D, Vallbona C. Patient exposition and provider explanation in routine interviews and hypertensive patients' blood pressure control. *Health Psychol.* 1987;6(1):29–42.
- 56 Kinmonth AL, Woodcock A, Griffin S, Spiegel N, Campbell MJ; The Diabetes Care From Diagnosis Research Team. Randomised controlled trial of patient centred care of diabetes in general practice: impact on current wellbeing and future disease risk. *BMJ.* 1998;317(7167):1202–1208.
- 57 Stewart M, Brown JB, Donner A, McWhinney IR, Oates J, Weston WW, et al. The impact of patient-centered care on outcomes. *J Fam Pract.* 2000;49(9):796–804.
- 58 Epstein RM. The science of patient-centered care. *J Fam Pract.* 2000;49(9):805–807.
- 59 Ong LM, de Haes JC, Hoos AM, Lammes FB. Doctor-patient communication: a review of the literature. *Soc Sci Med.* 1995;40(7):903–918.
- 60 Mead N, Bower P. Patient-centred consultations and outcomes in primary care: a review of the literature. *Patient Educ Couns.* 2002;48(1):51–61.
- 61 McCormack LA, Treiman K, Rupert D, Williams-Piehot A, Nadler E, Arora NK, et al. Measuring patient-centered communication in cancer care: a literature review and the development of a systematic approach. *Soc Sci Med.* 2011;72(7):1085–1095.
- 62 Rao JK, Anderson LA, Inui TS, Frankel RM. Communication interventions make a difference in conversations between physicians and patients: a systematic review of the evidence. *Med Care.* 2007;45(4):340–349.
- 63 Lewin SA, Skea ZC, Entwistle V, Zwarenstein M, Dick J. Interventions for providers to promote a patient-centred approach in clinical consultations. *Cochrane Database Syst Rev.* 2001(4):CD003267. doi:10.1002/14651858.CD003267.
- 64 Headache Study Group of the University of Western Ontario. Predictors of outcome in headache patients presenting to family physicians—a one year prospective study. *Headache.* 1986;26(6):285–294.
- 65 Starfield B, Wray C, Hess K, Gross R, Birk PS, D'Ugoff BC. The influence of patient-practitioner agreement on outcome of care. *Am J Public Health.* 1981;71(2):127–131.
- 66 Clever SL, Ford DE, Rubenstein LV, Rost KM, Meredith LS, Sherbourne CD, et al. Primary care patients' involvement in decision-making is associated with improvement in depression. *Med Care.* 2006;44(5):398–405.
- 67 Riccardi VM, Kurtz SM. *Communication and Counseling in Health Care.* Springfield, IL: Charles C Thomas Publisher Ltd; 1983.
- 68 Weiner SJ, Schwartz A, Sharma G, Binns-Calvey A, Ashley N, Kelly B, et al. Patient-centered decision making and health care outcomes: an observational study. *Ann Intern Med.* 2013;158(8):573–579.
- 69 Tamblyn R, Abrahamowicz M, Dauphinee D, Wenghofer E, Jacques A, Klass D, et al. Physician scores on a national clinical skills examination as predictors of complaints to medical regulatory authorities. *JAMA.* 2007;298(9):993–1001.
- 70 Levinson W, Roter D, Mullooly J, Dull V, Frankel R. Physician-patient communication: the relationship with malpractice claims among primary care physicians and surgeons. *JAMA.* 1997;277(7):553–559.
- 71 Ambady N, LaPlante D, Nguyen T, Rosenthal R, Chaumeton N, Levinson W. Surgeons' tone of voice: a clue to malpractice history. *Surgery.* 2002;132(1):5–9.
- 72 Beckman HB, Markakis KM, Suchman AL, Frankel RM. The doctor-patient relationship and malpractice: lessons from plaintiff depositions. *Arch Intern Med.* 1994;154(12):1365–1370.
- 73 Mazor KM, Roblin DW, Greene SM, Lemay CA, Firreno CL, Calvi J, et al. Toward patient-centered cancer care: patient perceptions of problematic events, impact, and response. *J Clin Oncol.* 2012;30(15):1784–1790.
- 74 Chewning B, Bylund CL, Shah B, Arora NK, Gueguen JA, Makoul G. Patient preferences for shared decisions: a systematic review. *Patient Educ Couns.* 2011;86(1):9–18.
- 75 Little P, Everitt H, Williamson I, Warner G, Moore M, Gould C, et al. Preferences of patients for patient centred approach to consultation in primary care: observational study. *BMJ.* 2001;322(7284):468–472.
- 76 Bendapudi NM, Berry LL, Frey KA, Parish JT, Rayburn WL. Patients' perspectives on ideal physician behaviors. *Mayo Clin Proc.* 2006;81(3):338–344.
- 77 Whitlock EP, Orleans CT, Pender N, Allan J. Evaluating primary care behavioral counseling interventions: an evidence-based approach. *Am J of Prev Med.* 2002;22(4):267–284.
- 78 Mazor K, Ockene J, Rogers H, Carlin M, Quirk M. The relationship between checklist scores on a communication OSCE and analogue patients' perceptions of communication. *Adv Health Sci Educ.* 2005;10(1):37–51.
- 79 National Alliance for Physician Competence. Guide to good medical practice-USA. September 22, 2008. <http://www.ama-assn.org/ama1/pub/upload/mm/377/ggmp-usa.pdf>. Accessed May 15, 2013.
- 80 de Haes H, Bensing J. Endpoints in medical communication research, proposing a framework of functions and outcomes. *Patient Educ Couns.* 2009;74(3):287–294.
- 81 Chandawarkar RY, Ruscher KA, Krajewski A, Garg M, Pfeiffer C, Singh R, et al. Pretraining and posttraining assessment of residents' performance in the fourth accreditation council for graduate medical education competency: patient communication skills. *Arch Surg.* 2011;146(8):916–921.
- 82 Larkin AC, Cahan MA, Whalen G, Hatem D, Starr S, Haley HL, et al. Human Emotion and Response in Surgery (HEARS): a simulation-based curriculum for communication skills, systems-based practice, and professionalism in surgical residency training. *J Am Coll Surg.* 2010;211(2):285–292.
- 83 Fallowfield L, Jenkins V, Farewell V, Saul J, Duffy A, Eves R. Efficacy of a Cancer Research UK communication skills training model for oncologists: a randomised controlled trial. *Lancet.* 2002;359(9307):650–656.
- 84 Mullen PD, Allen JD, Glanz K, Fernandez ME, Bowen DJ, Pruitt SL, et al. Measures used in studies of informed decision making about cancer screening: a systematic review. *Ann Behav Med.* 2006;32(3):188–201.
- 85 Saba GW, Wong ST, Schillinger D, Fernandez A, Somkin CP, Wilson CC, et al. Shared decision making and the experience of partnership in primary care. *Ann Fam Med.* 2006;4(1):54–62.
- 86 Thorne SE, Kuo M, Armstrong EA, McPherson G, Harris SR, Hislop TG. "Being known": patients' perspectives of the dynamics of human connection in cancer care. *Psychooncology.* 2005;14(10):887–898.
- 87 Glasgow RE, Emont S, Miller DC. Assessing delivery of the five "As" for patient-centered counseling. *Health Promot Int.* 2006;21(3):245–255.
- 88 Smith RC, Dwamena FC, Grover M, Coffey J, Frankel RM. Behaviorally defined patient-centered communication—a narrative review of the literature. *J Gen Intern Med.* 2011;26(2):185–191.

- 89 Schirmer JM, Mauksch L, Lang F, Marvel MK, Zoppi K, Epstein RM, et al. Assessing communication competence: a review of current tools. *Fam Med*. 2005;37(3):184–192.
- 90 Smith RC, Lyles JS, Mettler J, Stoffelmayr BE, Van Egeren LF, Marshall AA, et al. The effectiveness of intensive training for residents in interviewing: a randomized, controlled study. *Ann Intern Med*. 1998;128(2):118–126.
- 91 Smith RC, Marshall AA, Cohen-Cole SA. The efficacy of intensive biopsychosocial teaching programs for residents: a review of the literature and guidelines for teaching. *J Gen Intern Med*. 1994;9(7):390–396.
- 92 Fallowfield L, Jenkins V, Farewell V, Solis-Tripala I. Enduring impact of communication skills training: results of a 12-month follow-up. *Br J Cancer*. 2003;89(8):1445–1449.
- 93 Brown JB, Boles M, Mullooly JP, Levinson W. Effect of clinician communication skills training on patient satisfaction: a randomized, controlled trial. *Ann Intern Med*. 1999;131(11):822–829.
- 94 Stein T, Frankel RM, Krupat E. Enhancing clinician communication skills in a large healthcare organization: a longitudinal case study. *Patient Educ Couns*. 2005;58(1):4–12.
- 95 Laws MB, Epstein L, Lee Y, Rogers W, Beach MC, Wilson IB. The association of visit length and measures of patient-centered communication in HIV care: a mixed methods study. *Patient Educ Couns*. 2011;85(3):e183–e188. doi:10.1016/j.pec.2011.04.013.
- 96 American Academy on Communication in Healthcare. AACH website. <http://www.aachonline.org>. Accessed May 15, 2013.
- 97 Marcum ZA, Zheng Y, Perera S, Strotmeyer E, Newman AB, Simonsick EM, et al. Prevalence and correlates of self-reported medication non-adherence among older adults with coronary heart disease, diabetes mellitus, and/or hypertension [published online ahead of print January 7, 2013]. *Res Social Adm Pharm*. doi:10.1016/j.sapharm.2012.12.002.
- 98 Bolus R, Pitts J. Patient satisfaction: the indispensable outcome. *Manag Care*. 1999;8(4):24–28.
- 99 Swing SR. The ACGME outcome project: retrospective and prospective. *Med Teach*. 2007;29(7):648–654.
- 100 Nasca TJ, Philibert I, Brigham T, Flynn TC. The next GME accreditation system—rationale and benefits. *N Engl J Med*. 2012;366(11):1051–1056.
- 101 Swing SR, Beeson MS, Carraccio C, Coburn M, Iobst W, Selden NR, et al. Educational milestone development in the first 7 specialties to enter the Next Accreditation System. *J Grad Med Educ*. 2013;5(1):98–106.
- 102 Hulsman R, Ros W, Winnubst J, Bensing J. Teaching clinically experienced physicians communication skills: a review of evaluation studies. *Med Educ*. 2002;33(9):655–668.
- 103 Elsevier. *Patient Education and Counseling* website. <http://www.journals.elsevier.com/patient-education-and-counseling/>. Accessed May 15, 2013.
- 104 European Association for Communication in Healthcare. EACH website. <http://www.each.nl/>. Accessed May 15, 2013.
- 105 American Academy on Communication in Healthcare. *Medical Encounter* website. <http://www.aachonline.org/?page=MedicalEncounter>. Accessed July 10, 2013.